

# Entropy Regulariser Taxonomy and Gap Analysis

thesis-IV working notes (HyMeKo)

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## Abstract

We list every regulariser arm currently implemented in the spectral- entropy programme and place them on a taxonomy of orthogonal design axes. The exercise reveals six well-defined regularizer families that are absent from the current implementation but follow naturally from the same theoretical framework. Each gap is named, sketched, and ranked by likely empirical relevance.

## 1 Twelve currently implemented arms

| Arm                       | Formula                                                                       | Notes                                  |
|---------------------------|-------------------------------------------------------------------------------|----------------------------------------|
| baseline                  | none                                                                          | paired control                         |
| l2_weight_decay           | $\lambda \sum_l \ W_l\ _F^2$                                                  | classical $L^2$ control                |
| scalar_entropy            | $\lambda \cdot H(A)$                                                          | original Hajdu thesis-IV anchor        |
| kl_trajectory             | $\lambda \cdot D_{\text{KL}}(\hat{\lambda}^{t-1} \ \hat{\lambda}^t)$          | temporal regulariser, single layer     |
| scalar_entropy_normalized | $\lambda \cdot H(A) / \log_2 \text{rank}(A)$                                  | Path B; scale-invariant                |
| entropy_target            | $\lambda \cdot (H_{\text{norm}} - H^*)^2$                                     | Path A; IB-style attractor             |
| structural_composite      | $\lambda \cdot \max(H_{\text{norm}}, \sigma_{\text{max}}/2, 1 - \text{sr}/r)$ | Path C; BFT-bound terms                |
| entropy_unified           | $\lambda \cdot ((H_{\text{norm}} - H^*)^2 + \beta H_{\text{norm}})$           | A + B blend                            |
| entropy_adaptive          | $\lambda \cdot \exp(\eta H_{\text{norm}}) \cdot H_{\text{norm}}$              | H-acceleration feedback                |
| total_combined            | $\lambda \cdot (A + B + C)$                                                   | literal sum                            |
| entropy_target_ka         | $\lambda \cdot (H_{\text{norm}} - H_{KA}^*)^2$                                | KA-bound $H^*$                         |
| entropy_telgarsky         | $\lambda \sum_l (1 - l/L)^p H_{\text{norm}}^{(l)}$                            | per-layer, depth-decay                 |
| cross_layer_mi            | $\lambda \cdot \frac{1}{\binom{L}{2}} \sum_{i < j} I_2(\hat{K}_i; \hat{K}_j)$ | Sanchez-Giraldo MI on activation Grams |

## 2 Taxonomy axes

Each arm is a point in a multi-dimensional design space. Six axes are relevant.

### Axis 1 — Statistic source.

*Weight-side* (regulariser is a function of  $W$  only) vs *activation-side* (depends on a forward pass through data).

### Axis 2 — Spatial scope.

Per-layer (one regulariser term per layer) vs full-network adjacency (one term over the whole spectrum).

### Axis 3 — Pairwise interactions.

Marginal (single-layer or full-network entropy) vs pairwise (mutual information, joint entropy).

### Axis 4 — Temporal scope.

Single-step (current state only) vs trajectory (depends on history — KL, second derivative).

### Axis 5 — View aggregation.

Single view (dataflow OR factor) vs joint view (both views combined).

### Axis 6 — Functional shape.

Linear in entropy ( $\lambda H$ ); quadratic-target ( $\lambda(H-H^*)^2$ ); composite (max / sum of multiple terms); feedback / adaptive ( $\lambda$  depends on a derived quantity).

## 3 Coverage map

A matrix of (arm  $\times$  axis) showing where each lands. “•” means the arm uses that axis; blank means the alternative (or the axis is not exercised).

| arm                       | W | A | per-L | full-A | marg | MI | 1-step | traj | view-J | quad/comp |
|---------------------------|---|---|-------|--------|------|----|--------|------|--------|-----------|
| scalar_entropy            | • |   |       | •      | •    |    | •      |      |        |           |
| kl_trajectory             | • |   |       | •      | •    |    |        | •    |        |           |
| l2_weight_decay           | • |   |       | •      | •    |    | •      |      |        |           |
| scalar_entropy_normalized | • |   |       | •      | •    |    | •      |      |        |           |
| entropy_target            | • |   |       | •      | •    |    | •      |      |        | •         |
| structural_composite      | • |   |       | •      | •    |    | •      |      |        | •         |
| entropy_unified           | • |   |       | •      | •    |    | •      |      |        | •         |
| entropy_adaptive          | • |   |       | •      | •    |    |        | •    |        | •         |
| total_combined            | • |   |       | •      | •    |    | •      |      |        | •         |
| entropy_target_ka         | • |   |       | •      | •    |    | •      |      |        | •         |
| entropy_telgarsky         | • |   | •     |        | •    |    | •      |      |        |           |
| cross_layer_mi            |   | • | •     |        |      | •  | •      |      |        |           |

**Visible imbalance.** The implementation is heavily concentrated on {weight-side, full-network, marginal, single-step} regulariser shapes. Eleven of twelve arms are weight-side; eleven are full-network; eleven are marginal (no MI); eleven are single-step.

## 4 Six unfilled gaps

### 4.1 Gap A — Joint-view entropy regularisation

**What it would be.** Compute the trace-1 Gram matrices  $\hat{K}_{\text{dataflow}}$  and  $\hat{K}_{\text{factor}}$  from the two adjacency expansions of the same network and penalise their *joint* entropy:

$$\mathcal{L}_{\text{view}} = \lambda \cdot H_2(\hat{K}_{\text{dataflow}} \odot \hat{K}_{\text{factor}}).$$

Removes the user choice of view; the regulariser penalises whichever view (or combination) currently has highest joint entropy.

**Why interesting.** On every dataset where we tested both views, the magnitudes of the two view-results differed but never disagreed in sign. A joint-view regulariser would be at least as informative as either view alone, and would be *view-orientation-independent* (no longer needs the `--view` CLI flag).

**Effort.** 20 lines of code; reuses existing Hadamard primitive from `cross_layer_mi`.

### 4.2 Gap B — Class-aware entropy target ( $H_{\text{norm}}$ floor)

**What it would be.** The current  $H^* = 0.5$  default is fixed. Theory predicts the optimal  $H^*$  scales with the task’s intrinsic representational floor. For an  $n$ -class classification with adjacency rank  $r$ ,

$$H_{\text{class}}^* = \max(0.5, \log_2(n)/\log_2(r)).$$

This explains the EMNIST shrinkage we just observed (26 classes  $\Rightarrow$  floor  $\approx 0.48$ , but our fixed 0.5 is too aggressive).

**Why interesting.** Acts as a “hidden-variable correction” for the residual variation in our 7c results. If it fixes the EMNIST and ResMLP-20 underperformance, the universality claim becomes *architecture- and task-balanced-independent*.

**Effort.** 5 lines of code (already have the bookkeeping for `entropy_target_ka` which is similar). 2 hours of GPU time on the 4-dataset stress matrix.

### 4.3 Gap C — Temporal mutual information

**What it would be.** Apply the Sanchez-Giraldo Hadamard joint to the SAME layer at two consecutive time steps:

$$\mathcal{L}_{\text{temporal-MI}} = \lambda \cdot I_2(\hat{K}_l^{(t)}; \hat{K}_l^{(t-1)}).$$

The activation-side analogue of `kl_trajectory`, properly framed as “how much new information enters layer  $l$  per step”.

**Why interesting.** `kl_trajectory` was null on most datasets but gave a (later un-replicated) hint on ResMLP-20. A proper temporal MI version may capture the same effect more cleanly because it shares the activation-side framework with `cross_layer_mi`.

**Effort.** Need to cache previous-batch activations across reg- update steps. Memory cost:  $O(B \cdot d_l)$  per layer. 30 lines of code.

### 4.4 Gap D — Information bottleneck (full)

**What it would be.** Define  $\hat{K}_X$  from the input batch and  $\hat{K}_Y$  from labels. For hidden representation  $T = a_l(X)$ ,

$$\mathcal{L}_{\text{IB}} = \lambda \cdot I_2(\hat{K}_X; \hat{K}_T) - \beta \cdot I_2(\hat{K}_T; \hat{K}_Y).$$

This is the matrix-based information-bottleneck of Achille and Soatto.

**Why interesting.** The most rigorous theoretical framing in the deep-learning regularisation literature. Would let the paper say “our regulariser is the IB principle realised by the Sanchez-Giraldo estimator”.

**Effort.** 80 lines of code; needs label-side Gram, can double batch-forward cost. Plumbing through harness needs care.

### 4.5 Gap E — Per-layer entropy target

**What it would be.** Generalise `entropy_target` from a single  $H^*$  to a per-layer target  $H_l^*$ , allowing different layers to be regulated to different entropy levels. The depth schedule could be:

$$H_l^* = H_{\text{base}}^* \cdot \alpha^{(l-L/2)},$$

i.e. exponential decay or growth around the network middle.

**Why interesting.** Telgarsky’s depth-tradeoff result motivates preserving information in late layers. A per-layer target lets us encode that explicitly without the depth-weighted sum form (`entropy_telgarsky`) that broke on circles.

**Effort.** 15 lines of code. Hyperparameters:  $\alpha$  schedule.

### 4.6 Gap F — Hessian / Fisher entropy

**What it would be.** Penalise the Shannon entropy of the *Hessian* or *Fisher information matrix* eigenvalue distribution, not the weight adjacency:

$$\mathcal{L}_{\text{hess}} = \lambda \cdot H(\text{eigvals of } \nabla^2 \mathcal{L}).$$

Connects to the “flat minima” literature: lower Hessian-spectrum entropy means flatter minima, better generalisation (Hochreiter & Schmidhuber 1997).

**Why interesting.** The variance-reduction signature we observed empirically is theoretically tied to flat minima. Regulating Hessian entropy directly might be the *actual mechanism* for the -reduction effect.

**Effort.** Significant — Hessian computation is expensive ( $O(P^2)$  for  $P$  parameters); needs careful approximation (diagonal Fisher, low-rank Hutchinson estimator). 150 lines + numerics.

## 5 Ranking by likely impact

| Rank | Gap                    | Effort    | Hours of GPU | Likely return                                 |
|------|------------------------|-----------|--------------|-----------------------------------------------|
| 1    | B (class-aware $H^*$ ) | 5 lines   | 2            | high — fixes residual EMNIST/ResMLP shrinkage |
| 2    | A (joint-view entropy) | 20 lines  | 3            | medium — eliminates a knob                    |
| 3    | C (temporal MI)        | 30 lines  | 4            | medium — may revive the ResMLP-20 hint        |
| 4    | E (per-layer target)   | 15 lines  | 4            | medium — replaces the broken Telgarsky idea   |
| 5    | D (full IB)            | 80 lines  | 6            | high if it works, but expensive               |
| 6    | F (Hessian entropy)    | 150 lines | 12           | potentially high but engineering risk         |

## 6 Recommendation

**Implement gap B immediately** (class-aware  $H^*$ ). It is the hidden-variable correction directly suggested by the phase-7c data; 2-hour test on the stress matrix will confirm or refute. If it explains EMNIST and ResMLP-20, the universality claim is materially strengthened.

**Defer gaps A, C, E to a follow-up paper** or paper appendix. They round out the design space but don’t address the strongest empirical critique (the residual variance in normalised-entropy performance).

**Gap D (information bottleneck) deserves its own paper.** It is the most theoretically prestigious regulariser to derive in this framework but unrelated to the universality story; would dilute focus.

**Gap F (Hessian) is a different paper entirely.** The flat-minima connection is interesting, but Hessian regularisers are a substantial research thread of their own.

## 7 Diagram of the design space

| Statistic source       | Pairwise interactions | Temporal scope      |
|------------------------|-----------------------|---------------------|
| weight-side ● 11/12    | marginal ● 11/12      | single-step ● 11/12 |
| activation-side ● 1/12 | MI ● 1/12             | trajectory ● 2/12   |
| Spatial scope          | View aggregation      | Functional shape    |
| full-network ● 11/12   | single view ● 12/12   | linear ● 4/12       |
| per-layer ● 2/12       | joint view ● 0/12     | quad/comp ● 8/12    |

The framework is deeply over-developed in the upper-left of the design matrix (weight-side, full-network, marginal, single-step) and under-developed everywhere else. The activation-side framework opened up by `cross_layer_mi` (phase 8) is the most promising direction for future expansion — it unlocks gaps A, C, D, and E by *reusing the same Sanchez-Giraldo Hadamard primitive*.